

ADNAN KHAN

Operations Research | Stochastic Optimization | Industrial Engineering

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📍 Paris, France

📖 Research Profile

Master's student in Operations Research and Risk Analytics at CentraleSupélec, Université Paris-Saclay, with a Bachelor's in Industrial and Production Engineering and production internships in food processing and steel. I build optimization models for decisions under uncertainty, from two-stage stochastic programs with CVaR to routing, maintenance and capacity planning, implemented in Python with Gurobi.

🎓 Education

CentraleSupélec, Université Paris-Saclay

09/2025 to 12/2026

Research Master's (M2) in Operations Research and Risk Analytics, track Complex Systems Engineering

Paris, France

- Departmental scholarship holder.
- Coursework in Stochastic Optimization, Optimization of Complex Decisions, Decision Making, Data Analytics and Machine Learning, Reinforcement Learning, and Resilience of Critical Infrastructures.

NITER, University of Dhaka

04/2020 to 12/2024

B.Sc. in Industrial and Production Engineering (IPE)

Dhaka, Bangladesh

- CGPA 3.50 / 4.00, 87th percentile in the department. Merit-based scholarship.
- Coursework in Probability and Statistics, Operations Research, Operations Management, Supply Chain Management, Quality Control and Management, and Manufacturing Processes.

🔬 Research Experience

Master's Thesis: Optimizing Resilient Omnichannel Distribution Networks with Two-Stage Stochastic MILP and CVaR Objective

CentraleSupélec and EM Normandie Business School | Supervisor: Prof. Jakob Puchinger

Paris, France

- Built a two-stage stochastic MILP with CVaR that assigns every order a primary and a backup fulfillment node before disruptions, across home delivery, click-and-collect and store replenishment, with exact routing by route enumeration (3,573 variables, proven optimal in 3.4 s).
- Showed that pre-committed backups carry most of the value. They cut expected cost by 32.1% and unserved demand from 127 to 48 units, and planning with scenarios instead of their average saved up to 13.1%, mostly through the choice of backups.
- Validated the findings on generated instances with up to 42 demand points and 25 scenarios (optimal in 64 s with Gurobi), and out of sample on 300 new scenarios per instance, where backups still cut the objective by 8.2% and 20.0% (95% bootstrap intervals).

Research Intern, Paris Brain Institute (ICM)

06/2026 to Present

Supervisors: Dr. Delphine Oudiette and Borjan Milinkovic | Mathematical modeling and data analysis

Paris, France

- Built a data pipeline combining 60 Hz eye-tracking with 10 to 32 Hz keyboard data into 30 quantitative features spanning pupillometry, gaze and AOI, blink dynamics and typing behavior, across 32 participants.
- Handled the sampling-rate mismatch with windowed aggregation instead of resampling, choosing the window length by comparing between-window and within-window entropy.
- Applied k-means clustering and Gaussian Hidden Markov Models to infer latent attentional states, with permutation testing and a 22-seed robustness check on the resulting state structure.
- Designing a convergent-validity test linking the inferred states to post-block self-reported absorption.

B.Sc. Thesis: Differential Effects of Manufacturing Exports, Inflation, and Labor Productivity on Remittances

NITER, University of Dhaka | Supervisor: Oyshik Bhowmik

Dhaka, Bangladesh

- Panel study of remittance inflows to Bangladesh, India and Pakistan from 1990 to 2022, using panel quantile regression to separate effects under low and high remittance regimes rather than at the mean only.
- Established robustness through multicollinearity checks, panel unit root and cointegration tests, with ARIMA and SARIMAX imputation of missing series before estimation.

🔧 Optimization Projects

Capacity Planning and Machine Replacement under Uncertainty

03/2026 to 04/2026

Stochastic Optimization course, CentraleSupélec | Supervisor: Prof. Yiping Fang | L-shaped method, CVaR, MDPs

- Two-stage stochastic LP with recourse for capacity planning, solved both as a sample average approximation extensive form and by the L-shaped method, with the two cross-checked against each other, then extended to a risk-averse CVaR objective.

- Infinite-horizon MDP over ten deterioration states for machine replacement. Value iteration returned a control limit policy that keeps the machine in states 1 to 7 and replaces it from state 8, and a dual LP over discounted occupancy measures priced reliability at about 0.65 k€ per 1% output increase.

Optimization of Maintenance Planning for Offshore Wind Turbines12/2025

Optimization of Complex Decisions course, CentraleSupélec | Team project with Patricia Call Cerezo | Pyomo, HiGHS

- MILP scheduling preventive and corrective maintenance over a 30-week horizon under crew, vessel travel and maintenance-window constraints, minimizing preventive, corrective, transport and outage costs.
- At the given costs the optimal plan skips preventive maintenance, because an intervention costs 75 € against roughly 2 € per week of avoided failure cost and pays back only after 37.5 weeks, beyond the 30-week horizon. The model makes this break-even condition explicit.

Accelerated Life Testing and Warranty Optimization for LED Systems12/2025 to 01/2026

Reliability modeling, Weibull analysis, ALT, profit optimization

- Modeled accelerated lifetime and degradation data under temperature and current stress with right-censored Weibull distributions, then built Arrhenius and multi-stress ALT models to estimate MTTF, median life, B10 and warranty failure probabilities.
- Fed these into a warranty and pricing optimization model with demand sensitivity, which identified a strategy of long warranties and premium prices that maximizes expected profit at low warranty risk.

Industry Experience

Akij Food and Beverage Limited (AFBL), Production Intern, Bangladesh05/2024 to 07/2024

- Ran time studies across workstations to break down cycle time, identify bottlenecks and quantify capacity losses.
- Used process mapping and line balancing to design a future-state production layout and manpower allocation.

Kabir Steel Re-Rolling Mills (KSRM), Production Intern, Bangladesh05/2023 to 07/2023

- Applied fishbone and Pareto analysis to prioritize the main defect drivers and standardized corrective actions with a QC checklist.
- Defect rate fell 11% over the 27 days tracked after the corrective actions were introduced. Presented the findings to the production and quality teams.

Technical Skills

Optimization methods	MILP, two-stage stochastic programming with recourse, sample average approximation and out-of-sample evaluation, L-shaped method and Benders decomposition, CVaR and risk-averse objectives, duality and sensitivity analysis, Markov decision processes, vehicle routing, network design
Solvers and modeling	Gurobi, Pyomo, HiGHS
Programming and data	Python (NumPy, pandas, SciPy, scikit-learn, Matplotlib), C/C++ (basic), SQL (basic), LaTeX
Machine learning and statistics	Hidden Markov Models, k-means clustering, permutation tests, bootstrap intervals, Weibull and Kaplan-Meier survival analysis, panel regression

Certifications

- Operations Research Specialization, National Taiwan University (Coursera)
- McKinsey Forward Program, McKinsey & Company
- Career Essentials in Data Analysis, Microsoft and LinkedIn
- Data Science Bootcamp, AIQuest.org

Languages

English (IELTS 7.0, C1, November 2024) | French (B1) | Spanish (A2) | Bengali (native)

Honors and Competitions

- Top 10 teams, BIHRM Supply Chain Contest, 2023, with a 75% merit-based scholarship in the SCM program.
- Regionally selected and competed at national level, Bangladesh Mathematical Olympiad, 2017.

References

Dr. Delphine Oudiette, Research Director (Inserm), Team Leader, DreamTeam, Paris Brain Institute
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Prof. Yiping Fang, Head of the Risk, Resilience and Reliability team, CentraleSupélec, Université Paris-Saclay
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